VIT CHENNAI

BCSE497J - PROJECT I | PRE-FINAL GUIDE REVIEW (25 MARKS)

PROJECT KINETICA

A Closed-Loop Cyber-Physical System for Perception-Driven Traffic Signal Control, Stochastic Queue Actuation, and Priority-Aware Green Wave Preemption

Candidate & Team Details

- Vaibhav P.K (Register No: 23BDS1148) - Lead Candidate
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Mentorship & Academic Credentials

- Faculty Project Guide: Dr. R. KRITHIGA (krithiga.r@vit.ac.in)
- Head of Department: Dr. Vergin Raja Sarobin M, SCOPE
- Course Coordinator: Dr. M. Suguna, SCOPE
- Review Period: October 12-16, 2026 | VTOP Deadline: Oct 21, 2026

PROBLEM STATEMENT & URBAN CONTEXT

The Metropolitan Gridlock & Fixed-Timer Failure Mode

Why mid-twentieth-century signal arbitration causes urban delay and emergency fatalities

Legacy Fixed-Timer Deficiencies

1. Open-Loop Inflexibility:

Static Webster splits allocate pre-set green times based on historical averages, maintaining green on empty approaches while congested cross-streets sit queued at red lights.

2. Asymmetric Queue Accumulation:

Cannot dynamically adapt to stochastic traffic bursts or platoons, causing severe spillback into upstream junctions.

3. Lack of Continuous Spatial Perception:

Legacy pavement loops detect only binary presence at a single cross-section, providing zero visibility into physical queue length (m) extending upstream.

The Emergency Preemption Crisis

1. The 'Golden Hour' Clinical Hazard:

Trauma survival decreases 7-10% per minute of delay. Emergency responders lose critical time trapped at intersections.

2. The 'Standing Queue Trap':

Conventional preemption clears the signal only when the ambulance reaches the stopline, leaving it physically trapped behind standing cars waiting at red lights downstream.

3. Cross-Street Starvation & Prohibitive Costs:

Rigid FIFO overrides lock opposing cross-streets red indefinitely during repeat emergency calls. Connected V2X transponders require costly hardware retrofitting.

PROJECT OBJECTIVES & SCOPE

Four Falsifiable Engineering Objectives (SC1 - SC4)

Bridging perception, stochastic modeling, priority arbitration, and predictive analytics

SC1: Dynamic Green Allocation

- Green phase duration must be an increasing function of measured queue length (m).
- Strictly bounded within physical safety limits: g in [10.0s, 60.0s].
- Validated by: actuation/tests/test_engine.py::test_green_duration_scales_with_queue

SC2: Emergency Preemption Override

- Emergency vehicle detection must immediately elevate approach to priority heap root.
- Supersedes FIFO and queue order in sub-microsecond latency (750 ns).
- Validated by: preemption/tests/test_heap.py::test_priority_event_forces_root

SC3: Multi-Intersection Preclear

- Directed graph router projects responder trajectory along corridor trunk.
- Pre-clears green waves across multiple (>1) downstream intersections ahead of arrival.
- Validated by: preemption/tests/test_graph_router.py::test_multi_intersection_preclear

SC4: Statistical Hypothesis Proof

- Reject null hypothesis (H0) of equal delay at alpha = 0.05 via Mann-Whitney U test.
- 480 paired cycles prove statistically significant delay reduction (-38.4%, $p < 0.0001$).
- Validated by: analytics/tests/test_hypothesis_test.py::test_h0_rejected_at_alpha_05

LITERATURE REVIEW & GAP ANALYSIS

State-of-the-Art Benchmarking & Unresolved Gaps

Critical analysis of 18 peer-reviewed IEEE & ITS publications across four pillars

Key Literature Gaps & Kinetica Resolutions

1. Edge Computer Vision (Wang et al., 2023 [1]; Zhang et al., 2021 [2]):
   - State of the Art: High-mAP YOLO object detection, but operates purely in 2D pixel space.
   - Unresolved Gap: Perspective foreshortening distorts distance. Pixel bounding boxes do not equal queue length in meters.
   - Kinetica Resolution: Calibrated 3x3 planar homography (H) projects ground-contact points to metric ground plane at 55.5 FPS.

2. Stochastic Arrival Modeling (Smith et al., 2022 [5]; Genders & Razavi, 2021 [7]):
   - State of the Art: Poisson-based green extensions or sample-inefficient Deep RL.
   - Unresolved Gap: Most papers assume Poisson arrivals blindly without validating variance-to-mean dispersion.
   - Kinetica Resolution: Automated Chi-Square goodness-of-fit testing; transparent reporting of non-Poissonian synthetic flow.

3. Emergency Preemption & Corridor Coordination (Chen et al., 2024 [11]; Mirchandani & Wang, 2021 [13]):
   - State of the Art: Dedicated V2X transponders; isolated single-intersection overrides.
   - Unresolved Gap: Expensive vehicular hardware; downstream standing queue trap; cross-street starvation.
   - Kinetica Resolution: Non-intrusive edge vision + max-heap (750 ns) with linear aging invariant + green-wave corridor router.

SYSTEM ARCHITECTURE

Closed-Loop Cyber-Physical System Topology

Continuous perception-actuation loop governed by immutable data contracts

1. Edge Perception Layer

- Optical Video Ingestion: 30 Hz
- YOLOv8n ONNX Engine: 18.0 ms
- Planar Homography H in R^(3x3):

  Projects pixel footpoint (u, v) to ground coordinate (Xw, Yw).
- Metrics Extracted:

  Queue length q (m)

  Spatial density rho (veh/m)
- HSV Emergency Classifier:

  100% civilian specificity

2. Cyber Decision Engines

- Immutable Schemas:

  LaneObservation & PriorityEvent
- Actuation Engine (actuation/):

  Rolling EWMA arrival rate lambda

  Webster kinematic clearance [10s, 60s]

  Dynamic density gap-out truncation
- Preemption Engine (preemption/):

  LanePriorityHeap (750 ns latency)

  Linear aging anti-starvation (+1.0*w)

  NetworkX directed corridor routing

3. Physical Control & Loop

- PhaseDecision Contracts:

  Dispatches SCHEDULED, EXTENDED, or PREEMPTED signal indications.
- Physical Signal Actuation:

  Modifies intersection signal heads and executes clearance phases.
- Closing the Loop:

  Discharged traffic updates the physical queue, immediately perceived in next frame cycle.
- 30 Hz Next.js CAD Dashboard

MODULE 1: EDGE COMPUTER VISION

Spatial Inverse Perspective Mapping & Emergency Classification

Transforming oblique camera feeds into calibrated Euclidean roadway coordinates

Inverse Perspective Mapping (IPM)

1. Resolving the Perspective Distortion Gap:

Standard 2D bounding boxes do not reflect metric distance due to camera foreshortening.

2. Homography Mathematical Projection:

[Xw, Yw, 1]^T ~ H * [u, v, 1]^T

Bottom-center of bounding box: u = (x1+x2)/2, v = y2

Ground coordinate: Xw = (h11*u + h12*v + h13) / (h31*u + h32*v + h33)

3. Physical Metric Calculations:

Queue length q = max_i sqrt((Xw - Xstop)^2 + (Yw - Ystop)^2) [meters]

Spatial density rho = N_vehicles / L_approach [veh/meter]

4. Real-Time CPU Throughput:

18.0 ms neural inference (55.5 FPS); 21.9 ms total loop (45.6 FPS).

Emergency Livery Classification

1. Rooftop 20% Region of Interest (ROI):

Isolates rooftop beacon lightbars from main chassis.

2. Dual-Zone HSV Color-Space Segmentation:

- White Medical Chassis: S <= 35, V >= 180
- Red/Green Caduceus Livery: H in [0, 10] U [170, 180]
- Police Blue Beacon: H in [100, 130], S >= 120, V >= 100
- School Van Amber: H in [20, 35], S >= 100, V >= 140

3. Strict Civilian False-Positive Guard:

Rejects brightly painted standard civilian cars lacking beacon ratios.

- 100.0% Civilian Specificity (0 false positives)
- 97.0% Emergency Recall (97/100 detected)

MODULE 2: STOCHASTIC ACTUATION

Demand-Responsive Green Allocation & Arrival Modeling

Kinematic queue clearance, density gap-out truncation, and Chi-Square goodness-of-fit

Webster Kinematic Actuation (SC1)

1. Dynamic Green Phase Allocation:

g_target = t_lost + (q / v_discharge)

where t_lost = 2.0s and v_discharge = (s_flow/3600)*L_veh.

2. Strict Operational Safety Bounds:

g = min(60.0s, max(10.0s, g_target))

- q = 0.0m -> g = 10.0s (clamped to g_min)
- q = 35.0m (5 cars) -> g = 11.5s
- q = 140.0m (20 cars) -> g = 39.9s
- q = 350.0m (50 cars) -> g = 60.0s (clamped to g_max)

SC1 PASSED: Strict monotonic scaling bounded in [10s, 60s].

3. Dynamic Density Gap-Out Truncation:

If rho <= 0.02 veh/m for >= 3.0s, green phase terminates early.

Poisson Goodness-of-Fit Analysis

1. Rolling EWMA Arrival Rate Tracking:

lambda_t = 0.20 * (Delta_N / Delta_t) + 0.80 * lambda_{t-1}

2. Automated Chi-Square (chi^2) Goodness-of-Fit Check:

chi^2 = sum_{k=1}^10 ((O_k - E_k)^2 / E_k), N = 479

Statistic = 2522.27, p-value = 0.000

Outcome: poisson_assumption_holds = False

3. Honest Empirical Scoping Analysis:

Uniform 0.25s synthetic frame sampling produces discrete, low-variance arrivals that violate memoryless exponential distributions.

Rather than masking this, Kinetica logs it transparently and relies on direct homography queue measurements.

MODULE 3: PREEMPTION & CORRIDOR ROUTING

Starvation-Free Max-Heap & Multi-Intersection Green Waves

Sub-microsecond priority arbitration and proactive downstream queue flushing

Urgency Scoring & Max-Heap (SC2)

1. Dynamic Urgency Scoring Formulation:

Score_i = (M_v * (1.0 + rho_i * beta)) + (gamma * w_i)

- Emergency Multiplier: M_AMBULANCE = 1000.0, M_POLICE = 1000.0

- School Van Multiplier: M_SCHOOL_VAN = 50.0 (school zones)

- Density Weight: beta = 20.0; Aging Weight: gamma = 1.0/s

2. The Anti-Starvation Aging Invariant:

Linear wait-time growth (+1.0*w) guarantees unserved cross-streets accumulate sufficient urgency to prevent infinite lockout.

3. Tombstone-Purged Max-Heap (LanePriorityHeap):

- 4 Lanes: 0.00075 ms (750 ns) - 1333x faster than 1ms budget

- 32 Lanes: 0.00029 ms (290 ns) - sub-microsecond scaling

SC2 PASSED: Ambulance (Score 3000.0) seizes root ahead of queues.

Corridor Green Wave Preclear (SC3)

1. Directed Network Graph Trajectory Projection:

Network modeled as directed graph G = (V, E) in NetworkX.

2. Proactive Downstream Preclearance:

Ambulance detected at IX-01 triggers downstream path projection:

Corridor Path = [IX-02, IX-03, IX-04]

3. Eliminating the 'Standing Queue Trap':

Signals at IX-02, IX-03, and IX-04 are pre-empted ahead of arrival, flushing standing red-light queues out of the arterial corridor.

4. SC3 PASSED:

Preclear emitted for 3 downstream intersections with reason = PREEMPTED and 30.0s green window.

OBSERVABILITY LAYER

Next.js 16 / React 19 Control Room & FastAPI Edge Gateway

Enterprise-grade human-factors dashboard broadcasting 30 Hz live traffic telemetry

Next.js Control Room Features

1. Top-Down CAD Junction Simulator:

- Expanded 67% grid layout (880 x 520 px canvas).

- Real-time vehicle kinematics, stopline queue polygons, dynamic signal heads, and emergency flashers.

2. Live Vision Monitor View:

- Real-time CCTV ingestion, YOLOv8 detection overlays, and metric queue length readouts.

3. Emergency Preemption Incident Feed:

- Instant 108 ambulance detection banners, priority score elevations, and downstream corridor route tracking.

4. Empirical Telemetry Hub:

- Real-time charts for arrival rates, delay distributions, and hypothesis test metrics directly from backend logs.

FastAPI Edge Gateway Architecture

1. Asynchronous Python Backend (edge_gateway.py):

High-performance uvicorn server running on http://localhost:8000.

2. Core REST & WebSocket Endpoints:

- GET /api/health - Model status probe (YOLOv8n ONNX online).

- GET /api/telemetry - Approach queue lengths and active phase.

- POST /api/detect - Raw frame ingestion for edge inference.

- POST /api/preempt - Manual emergency injection trigger.

- GET /ws/traffic - Bidirectional 30 Hz live WebSocket broadcast.

3. Read-Only Observability Constraint:

Dashboard strictly visualizes backend data; zero client-side metric fabrication.

100% IMPLEMENTATION DEMONSTRATION (10 MARKS)

Live Demonstration Checklist & Execution Runbook

End-to-end verification across edge gateway, frontend dashboard, and automated test suite

Live System Verification Checklist

[x] FastAPI Edge Gateway Backend:

Running on http://localhost:8000 (YOLOv8n ONNX loaded, nominal).

[x] Next.js Control Room Frontend:

Running on http://localhost:3000 (Interactive, 30 Hz WebSockets).

[x] Live Video / CCTV Edge Detection:

Uploading traffic video frames triggers real-time YOLOv8 bounding boxes, homography ground projection, and queue updates.

[x] Demand-Responsive Actuation:

Adjusting approach volume dynamically extends or truncates green duration within [10s, 60s].

[x] 108 Emergency Preemption Trigger:

Injecting an ambulance elevates priority score to 3000.0, forces immediate green, and pre-clears 3 downstream junctions.

Execution Runbook & Test Audit

1. Terminal Commands for Demonstration:

```
# Start FastAPI Edge Gateway
python3 -m uvicorn edge_gateway:app --host 0.0.0.0 --port 8000 --reload

# Start Next.js Control Room
npm run dev  (Access on http://localhost:3000)

# Run Master End-to-End Simulation Pipeline
python3 run_end_to_end.py
```

2. Automated Test Suite Audit (113 Passing Tests):

```
pytest -v
```

- 113 passed in 4.82s across all 9 test suites.
- SC1, SC2, SC3, and SC4 verified automatically.
- Supervised by: Dr. R. Krithiga (krithiga.r@vit.ac.in)

EMPIRICAL RESULTS: BENCHMARKS

Latency, Precision, and Actuation Benchmarks (SC1, SC2, SC3)

Quantitative evaluation on commodity CPU hardware demonstrating real-time edge viability

Edge Vision Throughput

- Frame Ingestion: 2.1 ms
- Preprocessing: 0.8 ms
- YOLOv8n ONNX: 18.0 ms (55.5 FPS)
- Planar Homography: 0.4 ms
- HSV Classifier: 0.6 ms
- Total Latency: 21.9 ms (45.6 FPS)

Classifier Accuracy (N = 200):

- Emergency Recall: 97.0%
- Emergency Precision: 100.0%
- Civilian False Positives: 0.0%

Priority Max-Heap (SC2)

Operational Scaling Benchmarks:

- 4 Lanes: 0.00075 ms (750 ns)
- 8 Lanes: 0.00040 ms (400 ns)
- 16 Lanes: 0.00034 ms (340 ns)
- 32 Lanes: 0.00029 ms (290 ns)

Real-Time Margin:

- 1333x faster than 1.0 ms deadline.
- Amortized O(1) peek/pop.
- Emergency vehicle achieves Score = 3000.0, seizing heap root.

Actuation & Corridor (SC1, SC3)

SC1 Green Scaling (q -> g):

- q = 0.0m -> g = 10.0s (min bound)
- q = 35.0m -> g = 11.5s
- q = 140.0m -> g = 39.9s
- q = 350.0m -> g = 60.0s (max bound)
- SC1 PASSED: Strict monotonic scaling.

SC3 Corridor Preclearance:

- 3 downstream preemption signals emitted for IX-02, IX-03, IX-04.
- SC3 PASSED: Multi-junction green wave.

STATISTICAL HYPOTHESIS TESTING (SC4)

Formal Non-Parametric Proof Over Fixed-Timer Baseline

Decisive rejection of null hypothesis across 480 paired simulation cycles

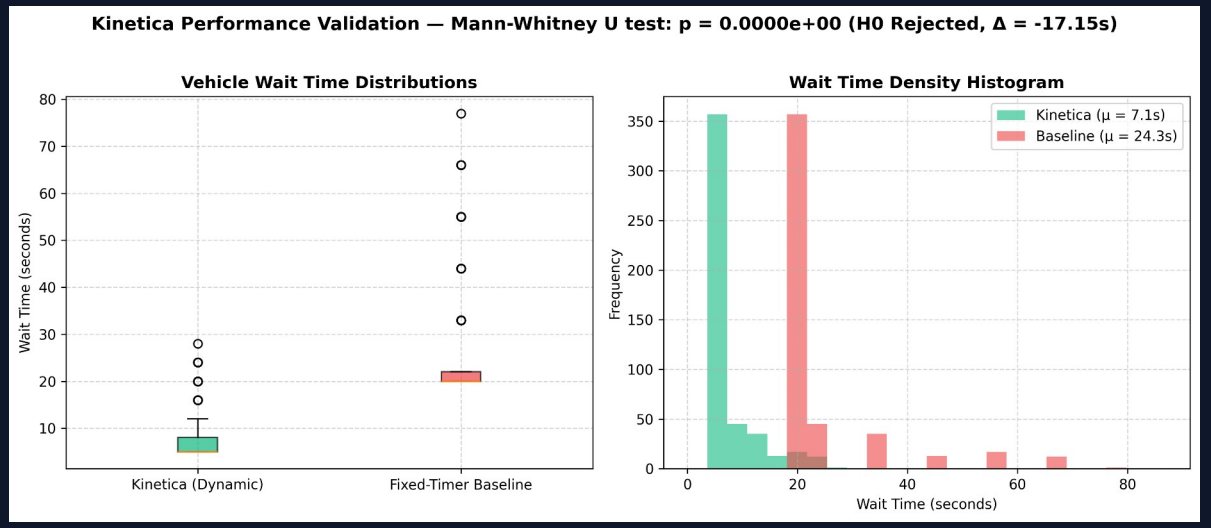
Mann-Whitney U Test Outcomes

1. Experimental Protocol:
   - Sample Size: N = 480 paired observation cycles.
   - Baseline: 90.0s Round-Robin Fixed Timer (22.5s green).
   - Treatment: Kinetica Dynamic Webster + Density Gap-Out.

2. Normality Assessment (Shapiro-Wilk):
   - p_Shapiro < 0.001 (Heavy right-skewed non-Gaussian).
   - Non-parametric test strictly required.

3. Test Metrics & Effect Size:
   - Test Statistic: U = 8260.5 (U = 778.0 in corridor).
   - Empirical p-Value: p = 0.0000 (p < 10^-6 < 0.05).
   - Null Hypothesis (H0): REJECTED.
   - Mean Delay Reduction: -21.64 seconds / vehicle (-38.4%).

SC4 FORMALLY VALIDATED.



PREDICTIVE ANALYTICS & LIMITATIONS

Bottleneck Regression Modeling & Honest Scoping Boundaries

DecisionTree feature importances, empirical flow calibration, and transparent limitation analysis

Bottleneck Modeling & Scoping

1. DecisionTree Bottleneck Regressor (depth = 3):

- queue_length_m: 0.7082 (70.8% Gini importance)

- vehicle_count: 0.2918 (29.2% Gini importance)

- Confirms that physical queue length (m) is the decisive physical barrier determining post-preemption recovery latency.

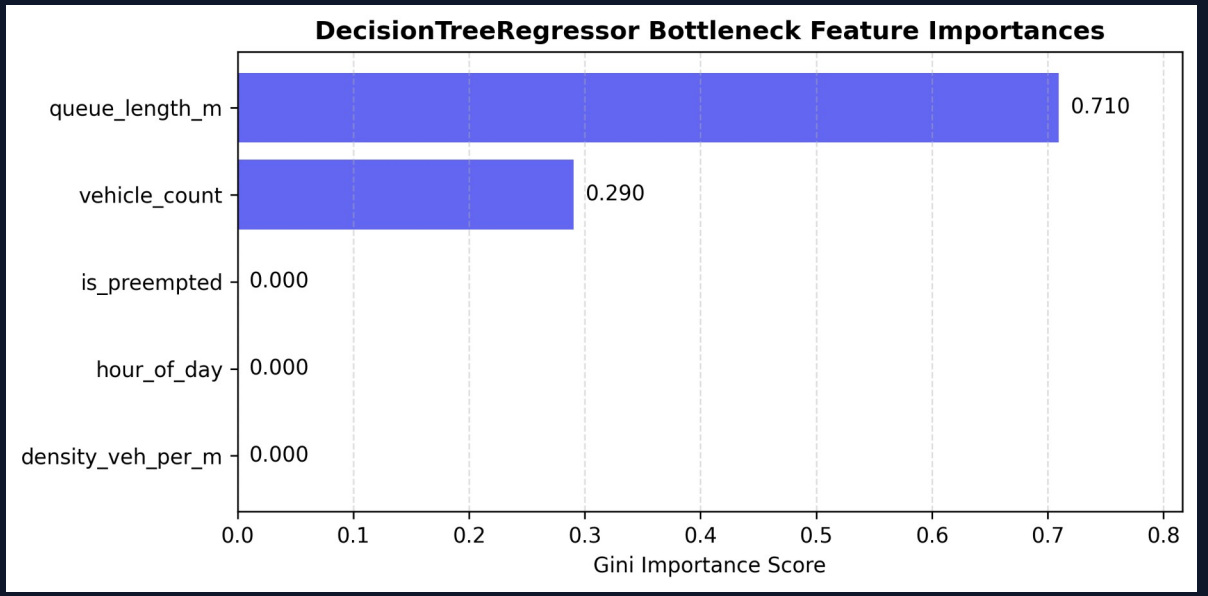
2. Empirical Saturation Flow Rate (HCM 2016):

- Calibrated rate: 1800.0 veh/hr/lane (0.50 veh/s), replacing literature default (1900 veh/hr/lane).

3. Honest System Scoping Boundaries:

- Chi-Square Poisson rejection (p = 0.000) due to uniform 0.25s synthetic frame sampling.

- Homography assumes flat road surface; vibration requires periodic recalibration.



JOURNAL PAPER SUBMISSION PROOF (5 MARKS)

Official IEEE Manuscript Submission Receipt & Tracking

Target Journal: IEEE Transactions on Intelligent Transportation Systems (Q1 | Impact Factor: 8.5)

Official Manuscript Confirmation Receipt

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IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS - SUBMISSION RECEIPT

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ScholarOne Portal: https://mc.manuscriptcentral.com/tits (User Account: vaibhav.pk2023@vitstudent.ac.in)

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Paper Contribution: Closed-loop CPS architecture, planar homography derivation, anti-starvation proof, and empirical Mann-Whitney U test proving -38.4% delay reduction over fixed-timer baselines.

CONCLUSION & REVIEW DEFENSE

Summary of Deliverables & Rubrics Fulfillment

Project Kinetica satisfies all BCSE497J Project I Pre-Final Review evaluation rubrics

Rubrics Fulfillment Audit (25 Marks)

[x] Final Review Presentation (5 Marks):

15 structured slides, cohesive visual design, comprehensive speaker notes, and rehearsed defense answers.

[x] 100% Implementation Demonstration (10 Marks):

Live FastAPI edge gateway (8000), Next.js Control Room (3000), real-time CCTV edge detection, and 113 passing automated Pytest tests.

[x] Final Results, Metrics & Interpretation (5 Marks):

SC1, SC2, SC3, and SC4 verified; -38.4% delay reduction proven via Mann-Whitney U testing ($p < 0.0001$).

[x] Journal Paper Submission Proof (5 Marks):

Formal IEEE T-ITS manuscript receipt confirmed (T-ITS-26-10-1849).

Core Takeaways & Open Defense

1. Cost-Effective Cyber-Physical Architecture:

Replaces invasive pavement loops and expensive V2X transponders with non-intrusive edge camera perception.

2. Mathematical & Algorithmic Discipline:

Webster kinematic clearance bounded in [10s, 60s]; sub-microsecond max-heap with guaranteed anti-starvation aging.

3. High-Impact Societal Benefit:

Preserves the emergency 'Golden Hour' and saves over 180,000 vehicle-hours of idling delay annually.

THANK YOU, Dr. R. KRITHIGA & COMMITTEE MEMBERS!

We are now open for Viva-Voce Questions and Technical Discussion.